Session 2: Solutions

Do not open until your write-up is done.

Main problem: Putnam 2017 A1

Answer. The positive integers not in S are exactly 1 and the multiples of 5.

Let $T = \{\, n \geq 2 : 5 \nmid n \,\}$. We prove $S = T$.

Step 1: $S \subseteq T$**.** T satisfies all three rules.

(a) $2 \in T$.

(b) If $n^2 \in T$, then $n^2 \geq 2$ so $n \geq 2$, and $5 \nmid n^2$ so $5 \nmid n$ (as 5 is prime). So $n \in T$.

(c) If $n \in T$, then $5 \nmid n+5$ and $(n+5)^2 \geq 49 \geq 2$. So $(n+5)^2 \in T$.

Since S is the smallest set satisfying (a), (b), (c), we get $S \subseteq T$.

This is the step people skip. "Smallest" is a hypothesis; it is used exactly here and nowhere else.

Step 2.

Lemma. If $n \in S$, then $n + 5 \in S$.

Proof. $n \in S$ gives $(n+5)^2 \in S$ by (c), hence $n + 5 \in S$ by (b). □

Step 3: $2, 3, 4, 6 \in S$**.**

$$
\begin{aligned}
2 \in S & \quad \text{by (a)} \\
49 = (2+5)^2 \in S, \text{ so } 7 \in S & \quad \text{by (c), then (b)} \\
64 = 49 + 15 \in S, \text{ so } 8 \in S & \quad \text{lemma three times, then (b)} \\
2916 = (49+5)^2 \in S & \quad \text{by (c)} \\
3136 = 2916 + 5 \cdot 44 = 56^2 \in S, \text{ so } 56 \in S & \quad \text{lemma, then (b)} \\
81 = 56 + 25 = 9^2 \in S, \text{ so } 9 \in S, \text{ so } 3 \in S & \quad \text{lemma, (b), (b)} \\
121 = 56 + 65 = 11^2 \in S, \text{ so } 11 \in S & \quad \text{lemma, (b)} \\
16 = 11 + 5 = 4^2 \in S, \text{ so } 4 \in S & \quad \text{lemma, (b)} \\
36 = 11 + 25 = 6^2 \in S, \text{ so } 6 \in S & \quad \text{lemma, (b)}
\end{aligned}
$$

Step 4: $T \subseteq S$**.** Every element of T is $\equiv 1, 2, 3$, or $4 \pmod 5$ and is at least 2. The smallest elements of T in those classes are $6, 2, 3, 4$ respectively, all in S by Step 3. By the lemma, every $n \in T$ is reached by adding 5 repeatedly to one of these. So $T \subseteq S$.

Combining Steps 1 and 4, $S = T$. The integers not in S are 1 and the multiples of 5. □

Grading notes.

- A write-up that only does Step 4 (shows lots of things *are* in S) with no Step 1 earns at most 2 or 3 points. The question asks which integers are *not* in S; without Step 1 you have not shown any integer is excluded.

- A write-up that only does Step 1 earns about 1 point.
- The lemma in Step 2 should be stated and proved separately. It turns Step 4 into a two-line argument instead of a mess.
- You do not need this exact chain in Step 3; any correct chain to $\{2, 3, 4, 6\}$ is fine. But every number you claim is in S must have a stated reason.

Optional problem: Putnam 2005 A1

Claim. Every positive integer n is a sum of numbers of the form $2^r 3^s$ $(r, s \geq 0)$ in which no summand divides another.

Proof. Strong induction on n. Base: $n = 1 = 2^0 3^0$.

Let $n \geq 2$ and assume the claim for all smaller positive integers.

Case n even. Write $n/2$ as such a sum and double every summand. Doubling preserves divisibility between summands ($a \mid b \iff 2a \mid 2b$), so no summand divides another.

Case n odd. Let 3^s be the largest power of 3 with $3^s \leq n$, so $3^s \leq n < 3^{s+1}$. Let $m = n - 3^s$. Then m is even and

$$0 \leq m < 3^{s+1} - 3^s = 2 \cdot 3^s.$$

If $m = 0$ then $n = 3^s$ and we are done. If $m > 0$, apply the hypothesis to $m/2$ and double, so m is a sum of terms $2^r 3^t$ with $r \geq 1$ and no term dividing another. Each such term is at most $m < 2 \cdot 3^s$, and since $r \geq 1$, $3^t \leq \text{term}/2 < 3^s$, so $t < s$.

Now $n = 3^s + (\text{those terms})$. Check divisibility:

- 3^s does not divide any term $2^r 3^t$, because $t < s$.
- No term $2^r 3^t$ divides 3^s, because $r \geq 1$.

So n is represented as required. □

Grading notes. The two bullets at the end *are* the problem. The inequality $t < s$ needs both the bound $m < 2 \cdot 3^s$ and $r \geq 1$; say both.